

DIGITAL IC DESIGN

The Digital IC Design track develops graduates into hardware engineering specialists capable of designing, verifying, and physically implementing complex microchips at a professional level. Across 1,435 hours, participants build hands-on expertise in hardware description languages, digital verification, FPGA prototyping, IC layout, and ASIC physical design, preparing them for high-demand roles across global semiconductor companies, automotive tech centers, and electronics R&D hubs.

JOB PROFILES

**Digital Design/RTL Engineer · Digital Verification Engineer ·
ASIC Physical Design Engineer · IC Layout Engineer ·
FPGA Engineer · CAD Engineer · QA Test Engineer**



TOTAL HOURS

1,398hrs

Theory + hands-on labs



DELIVERY

Blended

Online & on-site



UNIVERSITY GRADE

Good

Minimum grade required



Where graduates work

IC-Pedia AI-Chip Si-Vision GlobalFoundries Egypt OrionX
PyramidTech LLC Siemens EDA STMicroelectronics Mixel-Egypt
Analog Devices STMicroelectronics Goodix Technology Wasiela
Si-Ware Systems Pearl Semiconductor Qualcomm Intel



Who may apply

Open to graduates in Communications and Electronics Engineering and Computer Engineering with a minimum grade of Good or higher.



Selection process

- 01 IQ & English Exam**
💡 Online exam covering problem-solving aptitude and English proficiency.
- 02 Technical Exam**
💡 Computer-based assessment covering the program's core technical fundamentals.
- 03 Technical Interview**
💡 Interview with the technical committee.
- 04 Interpersonal Interview**
💡 Candidates who pass the technical interview are invited to this final stage.

Program content – 1,398 hours

CATEGORY	HOURS	SHARE	KEY COURSES
● Fundamental Courses	156 hrs	11%	System Administration, Introduction to Gen AI & Prompt Engineering, Software Fundamentals
● Core Courses	885 hrs	63%	Digital Design, Hardware Description Languages, Verification, Digital CMOS Circuit Design , FPGA Prototyping, IC Layout, ASIC Physical Design, Scripting
● Soft Skills	57 hrs	4%	Communication Essentials for Professionals, High Impact Presentations, Job Seeking Skills, Professional Demeanor
● Graduation Project	300 hrs	21%	

Hands-on labs are conducted in-person. Online components cover theory and self-study modules on MaharaTech.

Student Deliverable

- * All Course Assignments: Completion of all regular homework, practical tasks, and quizzes issued during each module.
- * Course Final Projects: A dedicated end-of-course project for each individual subject to test practical understanding.
- * Scholarship Graduation Project: The main, comprehensive final project that integrates everything learned throughout the scholarship, which must be completed and submitted to graduate

Before you apply

Explore these resources before your interview. Applicants are expected to have reviewed this material.

[Digital System Design using Verilog HDL](https://maharatech.gov.eg/course/view.php?id=2382)
https://maharatech.gov.eg/course/view.php?id=2382

[Mastering TCL Shell Scripting](https://maharatech.gov.eg/course/view.php?id=2278)
https://maharatech.gov.eg/course/view.php?id=2278

[Logic Circuits](https://youtube.com/playlist?list=PLkOpA9uAb9H3e0isTeEqkR-W5PwV8OSTG&si=_U-84vGEYy-6RCgA)
https://youtube.com/playlist?list=PLkOpA9uAb9H3e0isTeEqkR-W5PwV8OSTG&si=_U-84vGEYy-6RCgA

[ITI Values & Mission](https://iti.gov.eg/about-us)
https://iti.gov.eg/about-us

APPLY AT

ITI.GOV.EG

Applications open each intake period. Check the website for current dates and deadlines.